

Ram J. Zaveri

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SUMMARY

Ph.D. student in Neural Engineering with six years of research experience spanning machine learning, computer vision, and neural signal processing. My work combines biophysically grounded models of neural dynamics with quantitative analysis of electrophysiological and behavioral time series, investigating the principles underlying learning and adaptability in biological systems to build neuromimetic artificial systems that help design treatments for neurological disorders.

Research interests | Theoretical Neuroscience, Neural Signal Processing, Neuromorphic Systems, Machine Learning, Bayesian Brain Hypothesis, Predictive Coding, Continual Learning, Neural Basis of Learning

EDUCATION

Boston University, Boston, MA, USA Sep. 2025 – May 2029 (expected)

Ph.D. in Biomedical Engineering, Neural Engineering emphasis

West Virginia University, Morgantown, WV, USA May 2023 – May 2025

M.S. in Computer Science

GPA: 4.00/4.00

Thesis: *Efficient and Test-Time Adaptive Visual Object Tracking in the Wild*

West Virginia University, Morgantown, WV, USA Aug. 2019 – May 2023

B.S. (Honors) in Computer Science; Minor in Biology, Neuroscience emphasis

GPA: 4.00/4.00

Honors Foundations Scholar · Summa Cum Laude

Thesis: *Few-Shot Domain Adaptation for Cellular Instance Segmentation*

PUBLICATIONS

* denotes equal contribution.

Peer-Reviewed

- ▷ **R. J. Zaveri**, S. Patel, Y. Gu, and G. Doretto, "Improving Accuracy and Generalization for Efficient Visual Tracking," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 9468–9478. [\[link\]](#)
- ▷ S. Patel*, **R. J. Zaveri***, S. Chambers, Z. Randhawa, and G. Doretto, "Online Stochastic Optimization for Data with Temporal Dependencies," *European Conference on Computer Vision (ECCV) Workshops*, Springer, 2024, pp. 98–116. [\[link\]](#)
- ▷ **R. J. Zaveri***, V. Brume*, and G. Doretto, "Few-Shot Adaptation for Morphology-Independent Cell Instance Segmentation," *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 2024, pp. 1–5. [\[link\]](#)
- ▷ M. R. Keaton, **R. J. Zaveri**, and G. Doretto, "CellTranspose: Few-Shot Domain Adaptation for Cellular Instance Segmentation," *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2023, pp. 455–466. [\[link\]](#)
- ▷ M. R. Keaton, **R. J. Zaveri**, M. Kovur, C. Henderson, D. A. Adjero, and G. Doretto, "Fine-Grained Visual Classification of Plant Species in the Wild: Object Detection as a Reinforced Means of Attention," *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2021. [\[link\]](#)

RESEARCH EXPERIENCE

Graduate Fellow (PI: Dr. Frank Guenther) Mar. 2026 – Present
Speech Neuroscience Lab, Boston University Boston, MA, USA

Neural Representations of Impairment in Speech Motor Control | Developing an *in silico* model of stuttering to study how neural representations shape speech motor control in biological networks.

- ▷ Designed two cross-coupled continuous-time recurrent neural network (CT-RNN) sub-networks that jointly generate articulatory kinematic trajectories (electromagnetic articulography, EMA) and a voicing/glottal-source signal, coupling the two streams that must remain phase-aligned for fluent speech.
- ▷ Built the preprocessing chain for time-aligned articulatory-acoustic corpora: resampling and temporal alignment of EMA channels with the acoustic stream, low-pass filtering and detrending of articulator position traces, and extraction of voicing and pitch contours from the speech signal as supervision targets.
- ▷ Characterized the learned dynamics by analyzing latent-state trajectories and cross-network phase synchrony; systematically perturbing inter-network synchrony reproduces stuttering-like disfluencies (repetitions and blocks) in an otherwise fluent trained network, yielding a testable dynamical account of the disorder.

Research Rotations

Boston University

Sep. 2025 – Feb. 2026

Boston, MA, USA

Stangl Lab (PI: Dr. Matthias Stangl) — *Intracranial EEG signal processing* | Built an end-to-end processing and analysis pipeline for iEEG recorded from chronically implanted NeuroPace RNS depth electrodes in the medial temporal lobe.

- ▶ Implemented the full preprocessing chain: per-channel artifact and epileptiform-transient rejection, detrending, line-noise removal, and bandpass filtering within the limited recording bandwidth of the RNS system.
- ▶ Extracted theta-band ($\sim 4\text{--}12$ Hz) instantaneous amplitude and phase via time-frequency decomposition, and quantified oscillatory bouts against a fitted aperiodic ($1/f$) background rather than raw band power.
- ▶ Temporally synchronized and resampled neural time series against continuously sampled behavioral and positional streams, then applied circular-linear statistics to relate theta amplitude and phase to task variables such as distance to target and spatial location.

Natural Sounds and Neural Coding Lab (PI: Dr. Kamal Sen) — *Auditory scene analysis* | Co-designed a scalable cortical network model implementing top-down attentional modulation of inhibitory interneurons to suppress competing stimuli in an auditory spatial-localization task.

- ▶ Worked from a spatial-cue front end derived from binaural signals (interaural time and level differences) feeding excitatory-inhibitory cortical units, so that attentional gain acts on spatially indexed input channels.
- ▶ Analyzed model output using spike-timing-based measures rather than firing rate alone, evaluating how reliably a target stimulus remains decodable as competing sources are added across the spatial grid.
- ▶ Framed the model as a route toward biologically grounded sound-segregation algorithms for hearing-assistive devices operating in noisy, multi-talker environments.

Graduate Research Assistant (PI: Dr. Gianfranco Doretto)

Vision and Learning Group, West Virginia University

May 2023 – May 2025

Morgantown, WV, USA

Online Predictive Coding for Biomimetic Robotic Systems | Implemented an online learning framework for biomimetic robot control and vision grounded in the predictive coding theory of cortical function, learning hierarchical timescales through event demarcation to extract variable-length temporal representations from observations.

Online Learning for Time-Dependent Data | Designed an online stochastic optimization framework for self-supervised incremental learning on temporally correlated data streams; published at ECCV Workshops 2024.

Robust 2D Visual Tracking | Devised a test-time adaptation approach for single-object tracking under adverse visibility on edge devices, combining a backward-free dynamic adaptation scheme, new losses, and a CPU-efficient attention module for template-search correlation; published at WACV 2025.

Non-Convex Cell Segmentation | Developed a few-shot, domain-adaptive, morphology-independent cellular instance segmentation method for 2D volumes using specialized contrastive losses; published at ISBI 2024.

Undergraduate Research Assistant (PI: Dr. Gianfranco Doretto)

Vision and Learning Group, West Virginia University

Apr. 2020 – May 2023

Morgantown, WV, USA

Convex Cell Segmentation | Designed a few-shot domain-adaptive instance segmentation method for cells with convex morphology and a specialized contrastive loss for representation learning; published at WACV 2023 and selected for a Trainee Highlight Award at the 9th Annual BRAIN Initiative Meeting.

Plant Analysis | Developed a fine-grained visual classification method covering 1,000 plant species across five organs (leaf, flower, fruit, bark, high-density leaves), and released the accompanying long-tailed benchmark; published at CVPR Workshops 2021.

Fitbit Data Extraction and Digital Analysis | Built an IoT system for autonomous collection of wearable physiological time series (heart rate, activity, sleep staging), including cleaning, resampling, and windowed feature extraction from irregularly sampled streams, and trained models to infer target health outcomes. Part of the NSF Artificial Intelligence and Health 2020 workshop. [\[link\]](#)

INDUSTRY EXPERIENCE

Data Scientist, Predictive Maintenance

Sanofi Pasteur

Jun. 2025 – Aug. 2025

Swiftwater, PA, USA

- ▶ Analyzed continuous multivariate sensor telemetry from vaccine manufacturing filling lines in the Seeq advanced analytics platform, handling irregular sampling, sensor drift, and missing segments through resampling, alignment, and smoothing before modeling.

- ▷ Built a forecasting system with VAE-based LSTM networks that learns a latent representation of nominal machine behavior and flags deviations via reconstruction and prediction residuals, enabling anomaly detection without labeled fault data.
- ▷ Deployed ONNX models to the Seeq/Sanofi cloud for real-time streaming inference in production, supporting informed predictive maintenance decisions on live equipment.

Research Engineer (Joint contract with Vision and Learning Group) Jan. 2022 – May 2025
IstoVisio, Inc. (syGlass) — Microscopy Visualization & Quantification for VR Morgantown, WV, USA

AutoTrack — Adaptive 3D Cell Tracking & Tracing Jan. 2024 – May 2025

- ▷ Developed an efficient domain adaptation method for single-particle tracking in 3D time-series microscopy datasets.
- ▷ Designed a spatiotemporal fusion mechanism, framed as an attention module, to amplify task-relevant features during tracking.
- ▷ Partnered with the front-end team to integrate the feature into the shipping product.

syFind — Adaptive 3D Cell Counting & Segmentation Jan. 2022 – Dec. 2023

- ▷ Developed a 3D cell quantification and segmentation algorithm that adapts automatically to new datasets by performing adaptive representation learning on unseen microscopy data.
- ▷ Integrated the feature with the front-end team and demonstrated it with syGlass at the Society for Neuroscience (SfN) 2022 Expo.

TEACHING EXPERIENCE

Graduate Teaching Assistant Aug. 2023 – May 2025
West Virginia University Morgantown, WV, USA

CS 320 — Analysis of Algorithms (undergraduate; Spring 2024, Spring 2025) | Taught divide-and-conquer and shortest-path algorithms, dynamic programming, linear and integer programming, correctness proofs, and asymptotic space/time analysis. Held office hours, proctored exams, and graded assignments.

CS 678 — Computer Vision (graduate; Fall 2024) | Taught prerequisite mathematics (linear algebra, partial differential equations) and programming foundations (NumPy matrix and tensor operations); developed and graded assignments and held office hours.

CS 472 — Computer Graphics (undergraduate; Fall 2023) | Taught programming requirements (OpenCV, OpenGL, Pygame) and demonstrated an in-house 3D volumetric visualization system; assisted with course material and grading.

SCHOLARSHIPS, AWARDS & HONORS

Ph.D.

Neurophotonics Trainee Certificate, Boston University Neurophotonics Center 2026
 BU BME Graduate School Research Fellowship 2025 – 2026

Master's

NSF Research Traineeship (NRT), Graduate Trainee 2025
 Honorable Mention, WVU Inaugural AI Symposium 2025
 WVU Research Corporation Graduate Funding (\$22,000) 2023 – 2025
 Engineering Departmental Scholarship 2024
 Second Best Presentation Award, WVU Graduate Symposium 2024
 Upsilon Pi Epsilon Scholarship Award (\$1,000) 2024
 Trainee Highlight Award, 9th Annual BRAIN Initiative Meeting (top 60) 2023

Bachelor's

WVU Foundation Outstanding Senior (top 1% of ~4,000 graduating students) 2023
 Upsilon Pi Epsilon Dan Drew Award (highest distinction among ~600 nominees; \$4,000) 2023
 Honors Foundations Scholar · Summa Cum Laude 2023
 Office of Global Affairs Merit Award (one of ~900 students; \$14,000) 2021 – 2023
 Engineering Research Award (\$1,500) 2020 – 2023
 Level-1 Academic Scholarship (all semesters; \$7,000) 2019 – 2023
 Engineering Excellence Award (all semesters; \$1,500) 2019 – 2023
 President's List (all semesters) 2019 – 2023
 Upsilon Pi Epsilon Scholarship Award (\$1,500) 2022
 Arts & Sciences Departmental Award (\$2,000) 2021 – 2022

PRESENTATIONS

- ▷ *Improving Accuracy and Generalization for Efficient Visual Tracking*. WVU AI Symposium, 2025. **(Poster)**
- ▷ *Improving Accuracy and Generalization for Efficient Visual Tracking*. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2025. **(Poster)**
- ▷ *Few-Shot Adaptation for Morphology-Independent Cell Instance Segmentation*. IEEE International Symposium on Biomedical Imaging (ISBI), 2024. **(Poster)**
- ▷ *Few-Shot Adaptation for Morphology-Independent Cell Instance Segmentation*. WVU Graduate Symposium, 2024. **(Poster)**
- ▷ *Convex Cell Segmentation (extended work)*. 9th Annual BRAIN Initiative Meeting, 2023 — selected for the Trainee Highlight Award. **(Poster)**
- ▷ *CellTranspose: Few-Shot Domain Adaptation for Cellular Instance Segmentation*. IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023. **(Oral & Poster)**
- ▷ *Convex Cell Segmentation, with syGlass*. Society for Neuroscience (SfN), 2022. **(Exhibition)**
- ▷ *A Hierarchical Approach to Fine-Grained Visual Classification for Plant Image Analysis*. National Conference on Undergraduate Research (NCUR), 2021. **(Oral)**
- ▷ *A Hierarchical Approach to Fine-Grained Visual Classification for Plant Image Analysis*. Undergraduate Research Day at the Capitol (URDC), 2021. **(Oral)**
- ▷ *Fine-Grained Visual Classification for Plant Analysis*. WVU Undergraduate Research Symposium, 2020. **(Oral & Poster)**

SELECTED ACADEMIC & PERSONAL PROJECTS

Deriving Predictive Coding from Receptive-Field-Based Shunting Networks [\[link\]](#) | Designed a receptive-field-based shunting network with neurophysiologically valid dynamics and Hebbian-like learning for object recognition. Predictive coding emerged from the formulation, and formal derivation yielded a neurophysiologically plausible training procedure for deep networks.

Improving Cell Instance Segmentation with Test-Time Adaptation | Developed a test-time adaptation method for cellular instance segmentation on top of Cellpose, with novel losses inspired by Shannon entropy and self-supervised contrastive learning. Improved performance without additional labels or access to source data.

Biologically Inspired Continual Bayesian Learning [\[link\]](#) | Investigated a biologically plausible Bayesian learning framework for continual adaptability and reasoning under uncertainty. Designed an adaptive thresholding mechanism as a variational activation, mimicking leaky integrate-and-fire dynamics, and extended it to spiking neural networks for neuromorphic capability.

TECHNICAL SKILLS

Programming | Python, C/C++, Java, JavaScript, MATLAB, VHDL, PHP, HTML/CSS, SQL, $\LaTeX$

Libraries | PyTorch, TensorFlow, Keras, JAX, scikit-learn, TorchScript, ONNX, TorchIO, Napari, NumPy, SciPy, pandas, Matplotlib, OpenCV, OpenGL, Pygame

Signal Processing | Digital filtering (FIR/IIR, bandpass, notch, line-noise removal), resampling and multi-stream temporal alignment, artifact rejection, spectral estimation, time-frequency decomposition (wavelet, multitaper, Hilbert), aperiodic/oscillatory separation, phase-amplitude coupling, circular statistics, state-space and recurrent dynamical models, anomaly detection and forecasting on continuous sensor streams

Modalities | Intracranial EEG (NeuroPace RNS depth electrodes), EMG, electromagnetic articulography (EMA), speech acoustics, 3D time-series microscopy, wearable physiological sensors, industrial process telemetry

Tools | Linux, Git, Docker, Slurm, HPC, Bash, Selenium, Unity, Blender, Visual Studio, Android Studio, Seeq Advanced Analytics, Premiere Pro, After Effects

SELECTED COURSEWORK

Graduate | Graduate Linear Algebra & Statistics, Theoretical Linear Algebra, Dynamical Systems, Molecular Bioengineering, Neural Modeling, Neuromorphic Robotics Control & Modeling, Neural Networks, Computer Vision, Deep Learning, Advanced Analysis of Algorithms

Undergraduate | Multivariable Calculus, Applied Linear Algebra, The Living Cell, Neuroscience I & II, Sensory Neural

PROFESSIONAL SERVICE & SOCIETIES

Conference Reviewer | AAAI ('25, '26), CVPR ('24, '25), ICCV ('23, '25), ECCV ('24), ICLR ('24), MICCAI ('23, '24, '25), WACV ('23, '24, '25), ISBI ('24, '25), ICASSP ('24), BMVC ('23, '24)

Memberships

Institute of Electrical and Electronics Engineers (IEEE)	2022 – Present
Upsilon Pi Epsilon (UPE)	2021 – Present
Association for Computing Machinery (ACM)	2019 – Present

LEADERSHIP, MENTORSHIP & OUTREACH

Mentorship

2022 – 2025

- ▷ Mentored four students (S. Patel, D. Campa, C. Henderson, S. Chambers) through their research projects, developing their skills in problem solving, programming, presentation, and collaboration. All four presented their work at ECCV workshops or NCUR conferences.

Upsilon Pi Epsilon, WVU Chapter (*President*)

2021 – 2025

- ▷ Led the honor society for the computing and information disciplines; organized practical and theoretical workshops on computer architecture, explainable AI, and neural computation.

Motor Control Journal Club (*Member*)

2023 – 2025

- ▷ Presented and discussed neuroscience research relevant to prosthesis development, spanning computational neuroscience, computer science, and mechanical engineering.

Global InterVarsity (*Member*)

2019 – 2025

- ▷ Hosted weekly multicultural events and international dinners supporting students from diverse backgrounds.